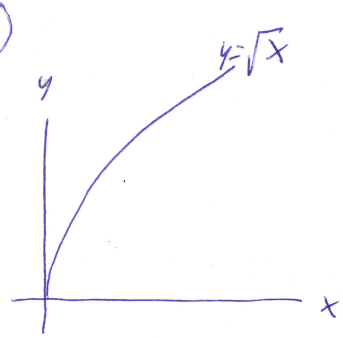
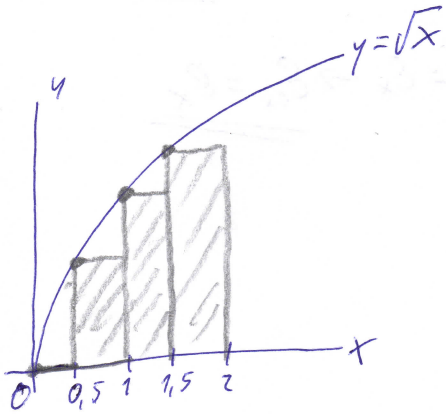
1



$n=4 \Rightarrow h=2$

a) l. čtverce

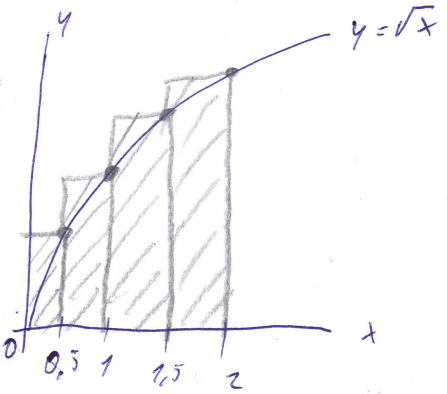


$$\int_0^2 \sqrt{x}\,dx \approx h\cdot(\sqrt{0}+\sqrt{0{,}5}+\sqrt{1}+\sqrt{1{,}5}) =$$

$$= 0{,}5\cdot\left(0+\sqrt{\tfrac{1}{2}}+1+\sqrt{\tfrac{3}{2}}\right) =$$

$$= 0{,}5\cdot\left(\frac{\sqrt{2}}{2}+1+\frac{\sqrt{6}}{2}\right) = \frac{\sqrt{2}}{4}+\frac{1}{2}+\frac{\sqrt{6}}{4}$$

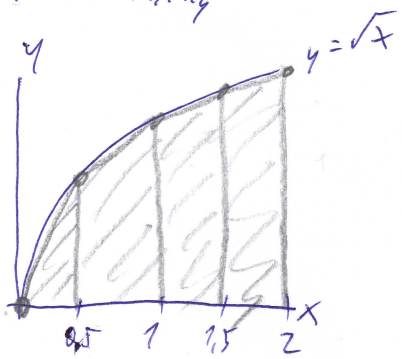
b) p. čtverce



$$\int_0^2 \sqrt{x}\,dx \approx h\cdot\left(\sqrt{\tfrac{1}{2}}+\sqrt{1}+\sqrt{\tfrac{3}{2}}+\sqrt{2}\right) = 0{,}5\left(\frac{\sqrt{2}}{2}+1+\frac{\sqrt{6}}{2}+\sqrt{2}\right) =$$

$$= \frac{\sqrt{2}}{4}+\frac{1}{2}+\frac{\sqrt{6}}{4}+\frac{\sqrt{2}}{2}$$

c) lichoběžníky



$$\int_0^2 \sqrt{x}\,dx \approx h\cdot\frac{\sqrt{0}+\sqrt{\tfrac{1}{2}}}{2} + h\cdot\frac{\sqrt{\tfrac{1}{2}}+\sqrt{1}}{2} + h\cdot\frac{\sqrt{1}+\sqrt{\tfrac{3}{2}}}{2} + h\cdot\frac{\sqrt{\tfrac{3}{2}}+\sqrt{2}}{2} =$$

$$= \frac{h}{2}\cdot\left(\sqrt{0}+2\underbrace{\sqrt{\tfrac{1}{2}}}_{\frac{\sqrt{2}}{2}}+2\sqrt{1}+2\underbrace{\sqrt{\tfrac{3}{2}}}_{\frac{\sqrt{6}}{2}}+\sqrt{2}\right) = \frac{1}{4}\left(\sqrt{2}+2+\sqrt{6}+\sqrt{2}\right) =$$

$$= \frac{\sqrt{2}}{4}+\frac{1}{2}+\frac{\sqrt{6}}{4}+\frac{\sqrt{2}}{4}$$

(2)

$E_x = x - \hat{x} \Rightarrow x = \hat{x} + E_x$

$\varepsilon_x = \frac{|E_x|}{|x|}$

$\cancel{E_{cx} = cx}$

$E_{cx} = cx - c\hat{x} = c(\hat{x} + E_x) - c\hat{x} = c\hat{x} + cE_x - c\hat{x} = cE_x \rightarrow \underline{\underline{E_{cx} = c \cdot E_x}}$

$\varepsilon_{cx} = \frac{|E_{cx}|}{|cx|} = \frac{|c \cdot E_x|}{|c(\hat{x} + E_x)|} = \left|\frac{E_x}{\hat{x} + E_x}\right| = \left|\frac{E_x}{x}\right| = \varepsilon_x \rightarrow \underline{\underline{\varepsilon_{cx} = \varepsilon_x}}$